

Operator Theory

Functional Analysis

Morales Carmona Ronny

Autumn 2026

Definition

Suppose V and W are Hilbert spaces and $T \in \mathcal{B}(V, W)$. The **adjoint** (or **hermitian conjugate**) of T , denoted by $T^\dagger$ is the map $T^\dagger : W \rightarrow V$ defined by

$$\langle Tf, g \rangle = \langle f, T^\dagger g \rangle$$

for every $f \in V$ and every $g \in W$.

Theorem

Suppose V and W are Hilbert spaces and $T \in \mathcal{B}(V, W)$. Then

$$T^\dagger \in \mathcal{B}(W, V), \quad (T^\dagger)^\dagger = T, \quad \text{and} \quad \|T^\dagger\| = \|T\|.$$

Theorem (Properties of $\dagger$)

Suppose V, W , and U are Hilbert spaces. Then

- i) $(S + T)^\dagger = S^\dagger + T^\dagger$ for all $T, S \in \mathcal{B}(V, W)$;
- ii) $(\alpha T)^\dagger = \alpha^* T^\dagger$ for all $\alpha \in \mathbb{C}$ and all $T \in \mathcal{B}(V, W)$;
- iii) $I^* = I$, where I is the identity operator on V ;
- iv) $(S \circ T)^\dagger = T^\dagger \circ S^\dagger$ for all $T \in \mathcal{B}(V, W)$ and $S \in \mathcal{B}(W, U)$.

Definition

An **operator** is an element of $\text{End}(V)$ for some linear space V .

Notation

The collection of bounded operators is denoted by $\mathcal{B}(V)$ instead of $\mathcal{B}(V, V)$.

Definition

An operator T on a linear space V is called **invertible** if T is bijective.

Theorem

A bounded operator T on a Hilbert space is invertible if and only if $T^\dagger$ is invertible. Furthermore, if T is invertible, then $(T^\dagger)^{-1} = (T^{-1})^\dagger$.

Theorem

Suppose U, V, W are normed spaces, $T \in \mathcal{B}(U, V)$, and $S \in \mathcal{B}(V, W)$. Then

$$\|ST\| \leq \|S\| \|T\|.$$

Theorem

If T is a bounded operator on a Banach space and $\|T\| < 1$, then $\text{Id} - T$ is invertible and

$$(I - T)^{-1} = \sum_{k \in \mathbb{N}} T^k.$$

Theorem

Suppose V is a Banach space. Then $\{T \in \mathcal{B}(V) : T \text{ is invertible}\}$ is an open subset of $\mathcal{B}(V)$.

Definition

Suppose T is a bounded operator on a Banach space V .

- i) T is called **left invertible** if there is some $S \in \mathcal{B}(V)$ such that $ST = \text{Id}$.
- ii) T is called **right invertible** if there is some $S \in \mathcal{B}(V)$ such that $TS = \text{Id}$.

Theorem

Suppose V is a Hilbert space and $T \in \mathcal{B}(V)$. Then the following are equivalent:

- i) T is left invertible.*
- ii) there exists $\alpha \in (0, \infty)$ such that $\|f\| \leq \alpha \|Tf\|$ for all $f \in V$.*
- iii) T is injective and has a closed range.*
- iv) $T^\dagger T$ is invertible.*

Theorem

Suppose V is a Hilbert space and $T \in \mathcal{B}(V)$. Then the following are equivalent:

- i) *T is right invertible.*
- ii) *T is surjective.*
- iii) *$TT^\dagger$ is invertible.*

Notation

From now on, the identity operator Id will be denoted simply by I .

Definition

Suppose T is a bounded operator on a Banach space V .

- A number $\alpha \in \mathbb{C}$ is called an **eigenvalue** of T if $T - \alpha I$ is not injective.
- A nonzero element $f \in V$ is called an **eigenvector** of T corresponding to an eigenvalue $\alpha \in \mathbb{C}$ if

$$Tf = \alpha f.$$

- The **spectrum** of T , denoted $\text{spec}(T)$, is defined as

$$\text{spec}(T) = \{\alpha \in \mathbb{C} : T - \alpha I \text{ is not invertible}\}.$$